Chapter 1

Introduction to signals and systems

1.1. Signals

Signal is defined as a set of information or data such as telephone or television signal. Generally the signals are functions of independent variable which is time. When an electrical charge is distributed over a surface, for example, the signal is the charge density, a function of space rather than time.

1.2. Classifications of signals

I. <u>Continous time and discrete time signals</u>

A signal x(t) is said to be continous one if it is defined over all values of t. Figure 1.1.a provides an example of continous time signal whose amplitude or value varies continously with time. These signals arises natureally when acoustic wave or light wave are converted to electrical signal. Conversion is done by transducer example include microphone and photocell. On the other hand, discrete time signals is defined only at discrete instants of time. Discrete time signal is often derived from continous time signal by sampling it at uniform rate. Let T_s denotes the sampling period and n denotes an integer that contain positive and negative values. Sampling continous time signal x(t) at time nT_s yields a sample value of (nT_s)

$$\mathbf{x(n)= x(nT_s)} \qquad \textbf{(1.1)}$$

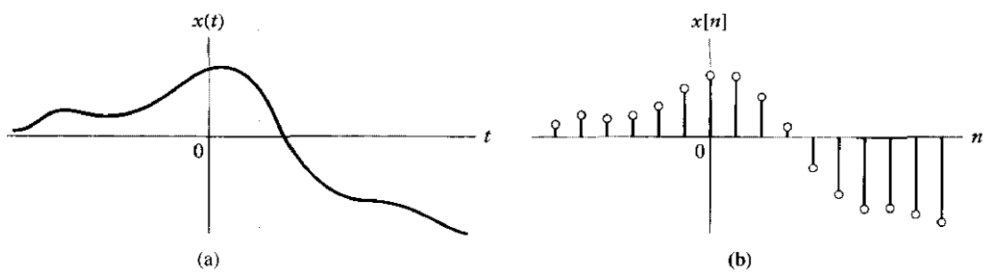


Figure 1.1 (a) Continues time signal x(t) (b) Representation of X(t) as discrete time value x(n)

II. <u>Even and Odd signals.</u>

A continuous-time signal $x(t)$ is said to be an *even signal* if it satisfies the condition

$$x(-t) = x(t) \quad \text{for all } t \qquad \textbf{(2.2)}$$

The signal $x(t)$ is said to be an *odd signal* if it satisfies the condition

$$x(-t) = -x(t) \quad \text{for all } t \qquad \textbf{(2.3)}$$

EXAMPLE (1.1) Develop the even/odd decomposition of a general signal $x(t)$ by applying the definitions of Eqs (1.2) and (1.3)

Solution: Let the signal $x(t)$ be expressed as the sum of two components $x_e(t)$ and $x_o(t)$ as follows:

$$x(t) = x_e(t) + x_o(t)$$

Define $x_e(t)$ to be even and $x_o(t)$ to be odd; that is,

$$x_e(-t) = x_e(t)$$

$$x_o(-t) = -x_o(t)$$

Putting $t = -t$ in the expression for $x(t)$, we may then write

$$\begin{aligned} x(-t) &= x_e(-t) + x_o(-t) \\ &= x_e(t) - x_o(t) \end{aligned}$$

Solving for $x_e(t)$ and $x_o(t)$, we thus obtain

$$x_e(t) = \frac{1}{2}\Big(x(t) + x(-t)\Big)$$

and

$$x_o(t) = \frac{1}{2}\Big(x(t) - x(-t)\Big)$$

III. Periodic and aperiodic signal

A *periodic signal* $x(t)$ is a function that satisfies the condition

$$x(t) = x(t + T) \quad \text{for all } t \tag{2.4}$$

where T is a positive constant. Clearly, if this condition is satisfied for $T = T_0$, say, then it is also satisfied for $T = 2T_0, 3T_0, 4T_0, \ldots$. The smallest value of T that satisfies Eq. (2.4) is called the *fundamental period* of $x(t)$. Accordingly, the fundamental period T defines the duration of one complete cycle of $x(t)$. The reciprocal of the fundamental period T is called the *fundamental frequency* of the periodic signal $x(t)$; it describes how frequently the periodic signal $x(t)$ repeats itself. We thus formally write

$$f = \frac{1}{T} \tag{2.5}$$

The frequency f is measured in hertz (Hz) or cycles per second. The *angular frequency*, measured in radians per second, is defined by

$$\omega = \frac{2\pi}{T} \tag{2.6}$$

Any signal $x(t)$ for which there is no value of T to satisfy the condition of Eq (2.4) is called an *aperiodic* or *nonperiodic signal.*

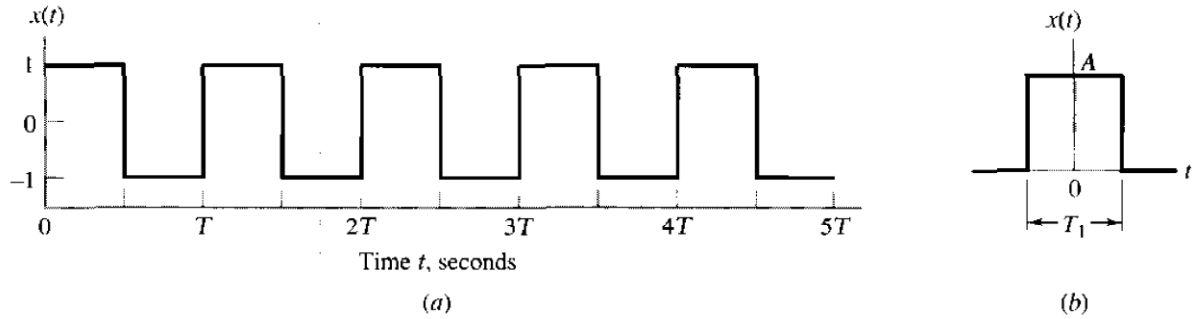


Figure 1.2 periodic rectangular pulse and a periodic one

IV. Deterministic and Random signal

A *deterministic signal* is a signal about which there is no uncertainty with respect to its value at any time. Accordingly, we find that deterministic signals may be modeled as completely specified functions of time. The square wave shown in Fig. **(2.2.a)** and the rectangular pulse shown in Fig. **(2.2.b)** are examples of deterministic signals.

On the other hand, a *random signal* is a signal about which there is uncertainty before its actual occurrence. Such a signal may be viewed as belonging to an ensemble or group of signals, with each signal in the ensemble having a different waveform. Moreover, each signal within the ensemble has a certain probability of occurrence. The ensemble of such signals is referred to as a *random process*. The *noise* generated in the amplifier of a radio or television receiver is an example of a random signal. Its amplitude fluctuates between positive and negative values in a completely random fashion. The EEG signal, is another example of random signal.

V. Energy signals, power signals

In electrical systems, a signal may represent a voltage or a current. Consider a voltage $v(t)$ developed across a resistor R, producing a current $i(t)$. The *instantaneous power* dissipated in this resistor is defined by

$$p(t) = \frac{v^2(t)}{R} \tag{2.7}$$

or, equivalently,

$$p(t) = Ri^2(t) \tag{2.8}$$

Based on this convention, we define the *total energy* of the continuous-time signal $x(t)$ as

$$\begin{aligned} E &= \lim_{T\to\infty} \int_{-T/2}^{T/2} x^2(t)\, dt \\ &= \int_{-\infty}^{\infty} x^2(t)\, dt \end{aligned} \tag{2.9}$$

and its *average power* as

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t)\, dt \tag{2.10}$$

From Eq (2.10) we readily see that the average power of a periodic signal $x(t)$ of fundamental period T is given by

$$P = \frac{1}{T} \int_{-T/2}^{T/2} x^2(t)\, dt \tag{2.11}$$

The square root of the average power P is called the *root mean-square* (rms) value of the signal $x(t)$.

In the case of a discrete-time signal $x[n]$, the integrals in Eqs. (1.9) and (2.10) are replaced by corresponding sums. Thus the total energy of $x[n]$ is defined by

$$E = \sum_{n=-\infty}^{\infty} x^2[n] \tag{2.12}$$

and its average power is defined by

$$P = \lim_{N\to\infty} \frac{1}{2N} \sum_{n=-N}^{N} x^2[n] \tag{2.13}$$

Here again we see from Eq (2.13) that the average power in a periodic signal $x[n]$ with fundamental period N is given by

$$P = \frac{1}{N} \sum_{n=0}^{N-1} x^2[n]$$

A signal is referred to as an *energy signal*, if and only if the total energy of the signal satisfies the condition

$$0 < E < \infty$$

On the other hand, it is referred to as a *power signal*, if and only if the average power of the signal satisfies the condition

$$0 < P < \infty$$

2.3. Basic Operations on signals

1. Operations performed on dependent variables

Amplitude scaling. Let $x(t)$ denote a continuous-time signal. The signal $y(t)$ resulting from amplitude scaling applied to $x(t)$ is defined by

$$y(t) = cx(t) \tag{2.14}$$

Addition. Let $x_1(t)$ and $x_2(t)$ denote a pair of continuous-time signals. The signal $y(t)$ obtained by the addition of $x_1(t)$ and $x_2(t)$ is defined by

$$y(t) = x_1(t) + x_2(t) \tag{2.15}$$

Multiplication. Let $x_1(t)$ and $x_2(t)$ denote a pair of continuous-time signals. The signal $y(t)$ resulting from the multiplication of $x_1(t)$ by $x_2(t)$ is defined by

$$y(t) = x_1(t)x_2(t) \tag{2.16}$$

That is, for each prescribed time t the value of $y(t)$ is given by the product of the corresponding values of $x_1(t)$ and $x_2(t)$. A physical example of $y(t)$ is an *AM radio signal*, in which $x_1(t)$ consists of an audio signal plus a dc component, and $x_2(t)$ consists of a sinusoidal signal called a carrier wave.

In a manner similar to Eq. **(2.16)** for discrete-time signals we write

$$y[n] = x_1[n]x_2[n]$$

Differentiation. Let $x(t)$ denote a continuous-time signal. The derivative of $x(t)$ with respect to time is defined by

$$y(t) = \frac{d}{dt}x(t) \tag{2.17}$$

For example, an *inductor* performs differentiation. Let $i(t)$ denote the current flowing through an inductor of inductance L, as shown in Fig **(2.3)** The voltage $v(t)$ developed across the inductor is defined by

$$v(t) = L\frac{d}{dt}i(t) \tag{2.18}$$

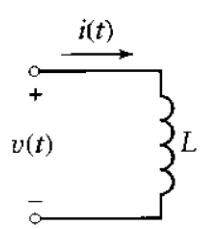


Figure (2.3) inductor with current i(t) and voltage v(t) across its terminals

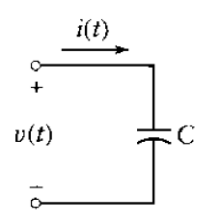


Figure (2.4) capacitor with voltage v(t) across its terminals, inducing current i(t)

Integration. Let $x(t)$ denote a continuous-time signal. The integral of $x(t)$ with respect to time t is defined by

$$y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau \tag{2.19}$$

where τ is the integration variable. For example, a capacitor performs integration. Let $i(t)$ denote the current flowing through a capacitor of capacitance C, as shown in Fig. **(2.4)** The voltage $v(t)$ developed across the capacitor is defined by

$$v(t) = \frac{1}{C}\int_{-\infty}^{t} i(\tau)\, d\tau \tag{2.20}$$

2. Operations performed on the independent variables

Time scaling. Let $x(t)$ denote a continuous-time signal. The signal $y(t)$ obtained by scaling the independent variable, time t, by a factor a is defined by

$$y(t) = x(at)$$

If $a > 1$, the signal $y(t)$ is a *compressed* version of $x(t)$. If, on the other hand, $0 < a < 1$, the signal $y(t)$ is an expanded (stretched) version of $x(t)$. These two operations are illustrated in Fig. **(2.5)**

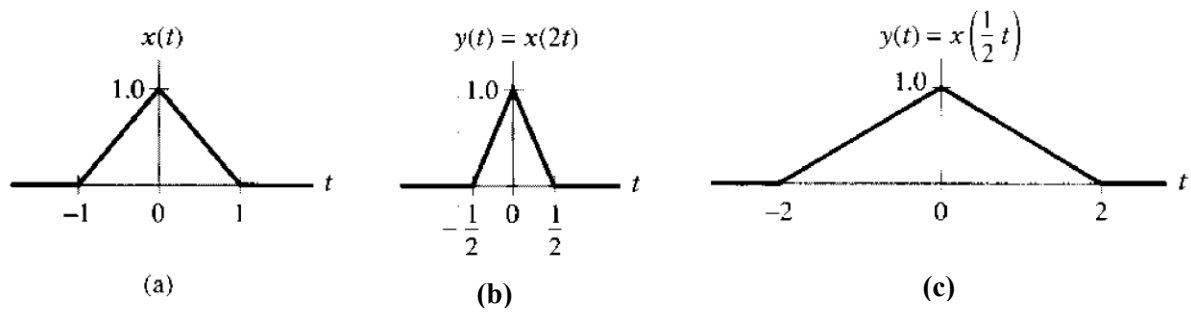


FIGURE (2.5) Time-scaling operation: (a) continuous-time signal $x(t)$, (b) compressed version of $x(t)$ by a factor of 2, and (c) expanded version of $x(t)$ by a factor of 2.

Reflection. Let $x(t)$ denote a continuous-time signal. Let $y(t)$ denote the signal obtained by replacing time t with $-t$, as shown by

$$y(t) = x(-t)$$

The signal $y(t)$ represents a reflected version of $x(t)$ about the amplitude axis.

The following two cases are of special interest:

- Even signals, for which we have $x(-t) = x(t)$ for all t; that is, an even signal is the same as its reflected version.
- Odd signals, for which we have $x(-t) = -x(t)$ for all t; that is, an odd signal is the negative of its reflected version.

Time shifting. Let $x(t)$ denote a continuous-time signal. The time-shifted version of $x(t)$ is defined by

$$y(t) = x(t - t_0)$$

where t_0 is the time shift. If $t_0 > 0$, the waveform representing $x(t)$ is shifted intact to the right, relative to the time axis. If $t_0 < 0$, it is shifted to the left.

Example (2.2):- Figure (2.6) shows a rectangular pulse x(t) of unit amplitude and unit duration. Find y(t)= x(t-2)

Solution: In this example, the time shift t_0 equals 2 time units. Hence, by shifting $x(t)$ to the right by 2 time units we get the rectangular pulse $y(t)$ shown in Fig. 1.22(b). The pulse $y(t)$ has exactly the same shape as the original pulse $x(t)$; it is merely shifted along the time axis.

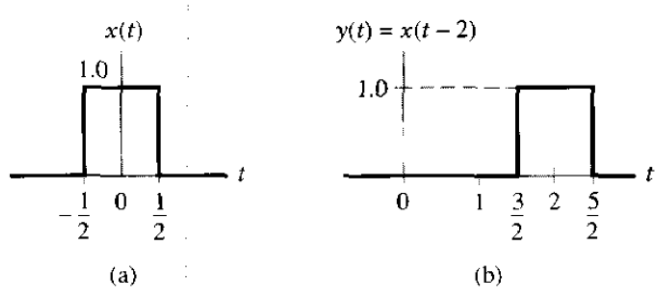


Figure (2.6) time shifting operation (a) continuous time signal in the form of rectangular pulse (b) time shifted version of x(t) by 2 time units

- **Periodic pulse for time shifting and scaling**

Let $y(t)$ denote a continuous-time signal that is derived from another continuous-time signal $x(t)$ through a combination of time shifting and time scaling, as described here:

$$y(t) = x(at - b) \tag{2.21}$$

This relation between $y(t)$ and $x(t)$ satisfies the following conditions:

$$y(0) = x(-b) \tag{2.22}$$

and

$$y\left(\frac{b}{a}\right) = x(0) \tag{2.23}$$

To correctly obtain $y(t)$ from $x(t)$, the time-shifting and time-scaling operations must be performed in the correct order. The proper order is based on the fact that the scaling operation always replaces t by at, while the time-shifting operation always replaces t by $t - b$. Hence the time-shifting operation is performed first on $x(t)$, resulting in an intermediate signal $v(t)$ defined by

$$v(t) = x(t - b)$$

Example (2.3):- Consider rectangular pulse x(t) of unit amplitude and duration of 2 time units shown in figure (2.7). Find y(t)=x(2t+3)

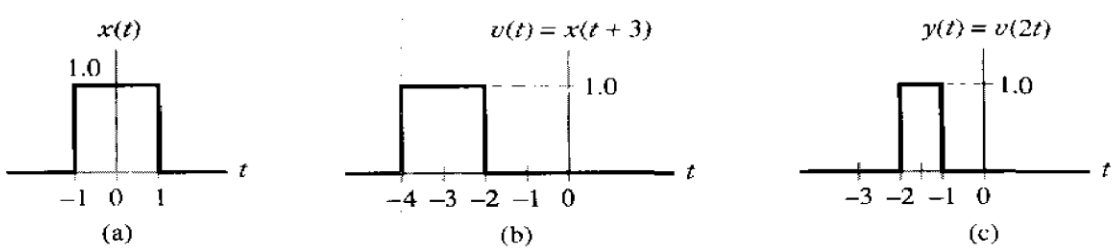


FIGURE (2.7) The proper order in which the operations of time scaling and time shifting should be applied for the case of a continuous-time signal. (a) Rectangular pulse $x(t)$ of amplitude 1.0 and duration 2.0, symmetric about the origin. (b) Intermediate pulse $v(t)$, representing time-shifted version of $x(t)$. (c) Desired signal $y(t)$, resulting from the compression of $v(t)$ by a factor of 2.

Solution: In this example, we have $a = 2$ and $b = -3$. Hence shifting the given pulse $x(t)$ to the left by 3 time units relative to the time axis gives the intermediate pulse $v(t)$ shown in Fig. 2.7 b). Finally, scaling the independent variable t in $v(t)$ by $a = 2$, we get the solution $y(t)$ shown in Fig. 2.7 (c).

2.4. Elementary signals

- **Exponential signals**

A real exponential signal, in its most general form, is written as

$$x(t) = Be^{at} \tag{2.24}$$

where both B and a are real parameters. The parameter B is the amplitude of the exponential signal measured at time $t = 0$. Depending on whether the other parameter a is positive or negative, we may identify two special cases:

- *Decaying exponential*, for which $a < 0$
- *Growing exponential*, for which $a > 0$

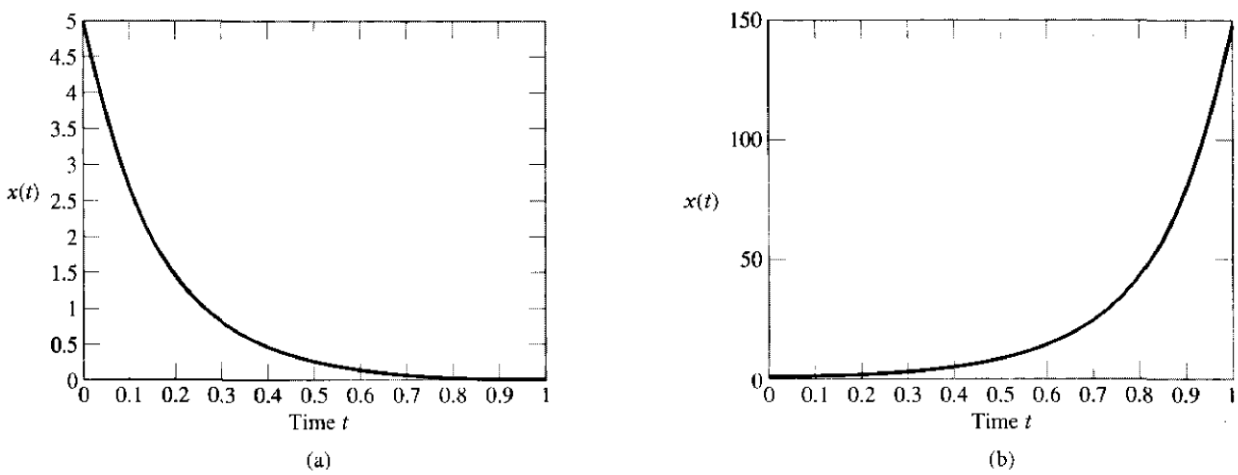


FIGURE (2.8) (a) Decaying exponential form of continuous-time signal. (b) Growing exponential form of continuous-time signal.

- **Sinusoidal signals**

The continuous-time version of a sinusoidal signal, in its most general form, may be written as

$$x(t) = A\cos(\omega t + \phi) \tag{2.25}$$

where A is the amplitude, ω is the frequency in radians per second, and ϕ is the phase angle in radians. Figure 1.29(a) presents the waveform of a sinusoidal signal for $A = 4$ and $\phi = +\pi/6$. A sinusoidal signal is an example of a periodic signal, the period of which is

$$T = \frac{2\pi}{\omega}$$

We may readily prove this property of a sinusoidal signal by using Eq **(2.25)** to write

$$\begin{aligned} x(t+T) &= A\cos(\omega(t+T)+\phi) \\ &= A\cos(\omega t+\omega T+\phi) \\ &= A\cos(\omega t+2\pi+\phi) \\ &= A\cos(\omega t+\phi) \\ &= x(t) \end{aligned}$$

Relationship between sinusoidal and complex exponential signals

Consider the complex exponential $e^{j\theta}$. Using *Euler's identity*, we may expand this term as

$$e^{j\theta} = \cos\theta + j\sin\theta \tag{2.26}$$

This result indicates that we may express the continuous-time sinusoidal signal of Eq. **(2.25)** as the real part of the complex exponential signal $Be^{j\omega t}$, where B is itself a complex quantity defined by

$$B = Ae^{j\phi} \tag{2.27}$$

That is, we may write

$$A\cos(\omega t+\phi) = \mathrm{Re}\{Be^{j\omega t}\} \tag{2.28}$$

where Re{ } denotes the real part of the complex quantity enclosed inside the braces. We may readily prove this relation by noting that

$$\begin{aligned} Be^{j\omega t} &= Ae^{j\phi}e^{j\omega t} \\ &= Ae^{j(\omega t+\phi)} \\ &= A\cos(\omega t+\phi) + jA\sin(\omega t+\phi) \end{aligned}$$

from which Eq **(2.28)** follows immediately. The sinusoidal signal of Eq. **(2.25)** defined in terms of a cosine function. Of course, we may also define a continuous-time sinusoidal signal in terms of a sine function, as shown by

$$x(t) = A\sin(\omega t+\phi) \tag{2.29}$$

which is represented by the imaginary part of the complex exponential signal $Be^{j\omega t}$. That is, we may write

$$A\sin(\omega t+\phi) = \mathrm{Im}\{Be^{j\omega t}\} \tag{2.30}$$

- **Exponentially damped sinusoidal signals**

The multiplication of a sinusoidal signal by a real-valued decaying exponential signal results in a new signal referred to as an exponentially damped sinusoidal signal. Specifically, multiplying the continuous-time sinusoidal signal $A\sin(\omega t + \phi)$ by the exponential $e^{-\alpha t}$ results in the exponentially damped sinusoidal signal

$$x(t) = Ae^{-\alpha t}\sin(\omega t + \phi), \qquad \alpha > 0 \tag{2.31}$$

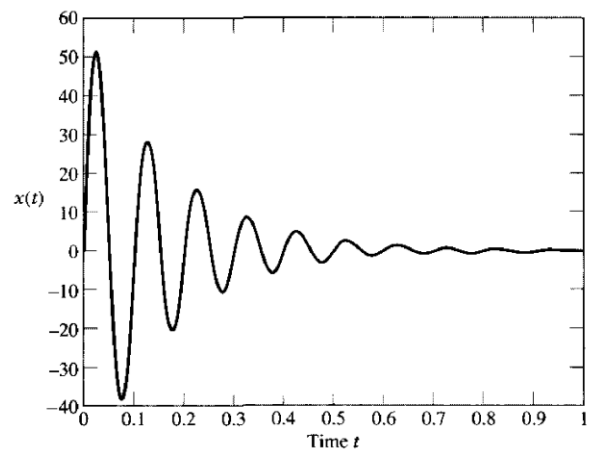


FIGURE (2.9) Exponentially damped sinusoidal signal $e^{-\alpha t}\sin\omega t$, with $\alpha > 0$.

- **Step function**

The discrete-time version of the step function, commonly denoted by $u[n]$, is defined by

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{2.32}$$

which is illustrated in Fig. **(2.10)**

The continuous-time version of the step function, commonly denoted by $u(t)$, is defined by

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases} \tag{2.33}$$

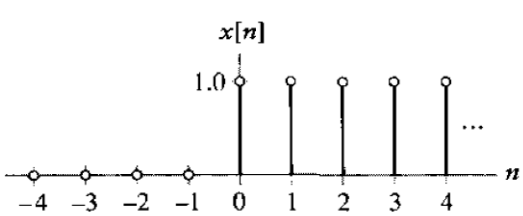


Figure 2.10 Discrete time version of step function

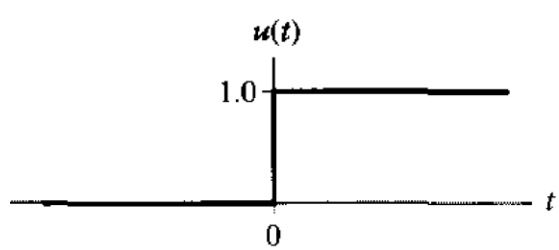


Figure 2.11 continuous time version of step function

- **Impulse Function**

The continuous-time version of the unit impulse, commonly denoted by $\delta(t)$, is defined by the following pair of relations:

$$\delta(t) = 0 \quad \text{for } t \neq 0 \tag{2.34}$$

and

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1 \tag{2.35}$$

Equation (2.34) says that the impulse $\delta(t)$ is zero everywhere except at the origin. Equation (2.35) says that the total area under the unit impulse is unity. The impulse $\delta(t)$ is also referred to as the *Dirac delta function*. Note that the impulse $\delta(t)$ is the derivative of the step function $u(t)$ with respect to time t. Conversely, the step function $u(t)$ is the integral of the impulse $\delta(t)$ with respect to time t.

$$\delta(-t) = \delta(t) \tag{2.36}$$

$$\int_{-\infty}^{\infty} x(t)\delta(t - t_0)\, dt = x(t_0) \tag{2.37}$$

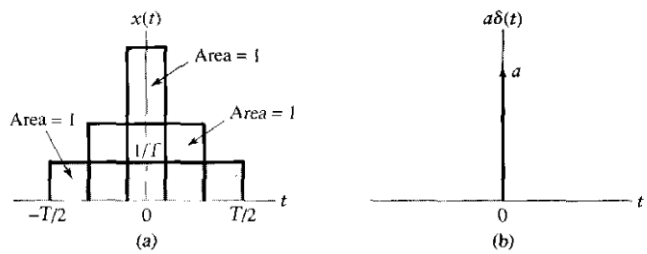


FIGURE (2.12) (a) Evolution of a rectangular pulse of unit area into an impulse of unit strength. (b) Graphical symbol for an impulse of strength a.

- **Ramp Function**

The impulse function $\delta(t)$ is the derivative of the step function $u(t)$ with respect to time. By the same token, the integral of the step function $u(t)$ is a ramp function of unit slope. This latter test signal is commonly denoted by $r(t)$, which is formally defined as follows:

$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases} \tag{2.38}$$

Equivalently, we may write

$$r(t) = tu(t) \tag{2.39}$$

The ramp function $r(t)$ is shown graphically in Fig. (2.13)

In mechanical terms, a ramp function may be visualized as follows. If the input variable is represented as the angular displacement of a shaft, then the constant-speed rotation of the shaft provides a representation of the ramp function. As a test signal, the

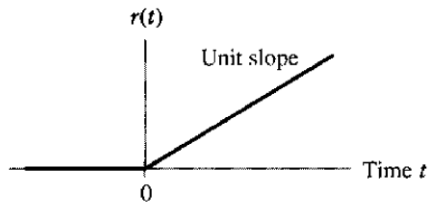


Figure 2.13 Ramp function with unit slope

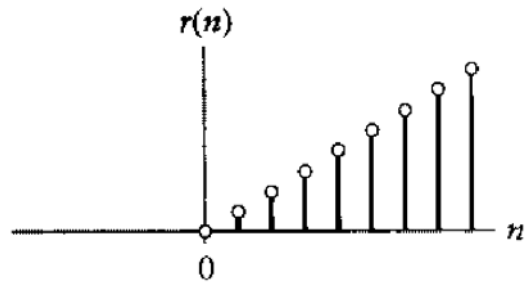


Figure 2.14 Discrete version of ramp function

ramp function enables us to evaluate how a continuous-time system would respond to a signal that increases linearly with time.

The discrete-time version of the ramp function is defined by

$$r[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{2.40}$$

or, equivalently,

$$r[n] = nu[n] \tag{2.41}$$

2.5. What is a System?

A system is an arrangement of physical components connected in such a manners to our desired objective-(to our liking). Electrical system is an electric circuit, the resistors, inductors, capacitors, transistors, and so forth that make up the circuit are the system components. A system is a two port network, one part is called the input port, and the other is called the output port, Thus if an excitation; or input signal is applied to the input terminals, a unique response or. output signal will be measurable or observable at the output terminals . A system that transforms a signal from one form to another form is called .a transducer. The relationship between the input (excitation cause) signal and the output (response , effect) signal is essential in defining (classification) a system . The following are examples of a system:-

- In an automatic speaker recognition system, the input signal is a speech (voice) signal, the system is a computer, and the output signal is the identity of the speaker.
- In a communication system, the input signal could be a speech signal or computer data, the system itself is made up of the combination of a transmitter, channel, and receiver, and the output signal is an estimate of the original message signal.
- In an aircraft landing system, the input signal is the desired position of the aircraft relative to the runway, the system is the aircraft, and the output signal is a correction to the lateral position of the aircraft.

Figure 2.15 Block diagram representation of a system

2.6. System classification?

System can be classified as follows:-

Classification aspect	variant
1- System type	Electrical, Electronic, Mechanical, Thermal, Hydrolic, Chemical, bio Engineering, and others
2- System categories	Open loop and closed loop systems
3- System operation	Continuous data- discrete data
4- System character	Linear, nonlinear- Causal, noncausal, static, dynamic
5- Parameters behavior in time	Time invariant, Time varying (parameters are function of time)
6- Number of variables (signals)	Single input single output SISO or MIMO
7- System signal characteristics	Deterministic, random or stochastic
8- Problem to be solved	Software, hardware
9- Computer in control	OFF line (computer outside control loop) ON line (computer inside control loop)

2.6.1 Causal and noncausal systems

Causal system means physical (nonanticipatory) system . A system is causal if its present response (output) to an input signal does not depends on a future values of that input; i.e. $y(t) = 0$ for $t<0$. This means that the output of the causal system at a given time depends on the present and past input signal; i.e. $y(t) = f\,[x(t-\lambda), x(t)]$. Remember that the future values of input signal is $x(t+\lambda)$ and the past values of input signal is $x(t-\lambda)$.

If the system not satisfying the conditions, then the system is non-causal; i.e. the noncausal system is anticipatory (non- physical) system. This means that the output (response) of the non causal system depends on the future values of the input signal ; i.e. $y(t) = f\,[x(t+\lambda)]$ at the present time. If the system impulse response is $h(t)$ then for causal system $h(t) = 0$ for $t<$. 0 and for noncausal system $h(t) = 0$ for $t< 0$.

$x(t) = 0$, for $t< 0$, for causal signal $\qquad$ $h(t) = 0$, for $t< 0$, for causal system
$x(t) \neq 0$, for $t < 0$, for noncausal signal $\qquad$ $h(t) \neq 0$, for $t< 0$, for noncausal system

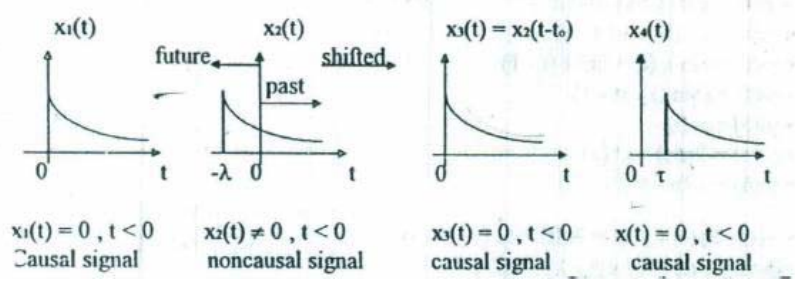


Figure 2.16 Causal and non causal signal examples

2.6.2 Linear and non linear systems

A system is linear if the property of superposition holds (satisfies)

- **Superposition property**:

let x(t) is the input signal of the system and y (t) its output signal, if the input signal is a $\alpha_1 X_1(t)$ then the output signal is $\boldsymbol{\alpha_1 y_1(t)}$, and if $\boldsymbol{\alpha_2 x_2(t)}$ is another input signal , then the output signal is $a_2 y_2(t)$ For superposition property holds ; i.e. if the input is $\boldsymbol{\alpha_1 x_1 + \alpha_2 x_2}$ then the output must be $\boldsymbol{\alpha_1 y_1 + \alpha_2 y_2}$ for linear system as shown in figure 2.17.

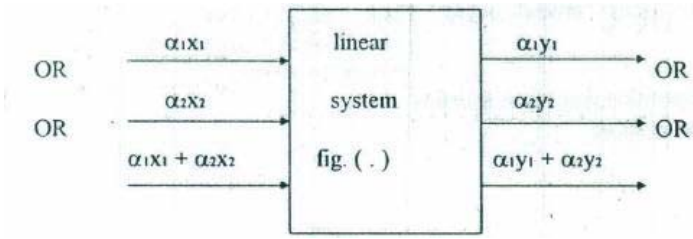


Figure 2.17 Linear system

Then the mathematical definition of linearity is given by:

$$\boldsymbol{if}\ x(t) = \sum_{i=0}^{n} \alpha_i x_i\,(t) = \textbf{input signal} \quad then\ \ y(t) = \sum_{i=0}^{n} \alpha_i x_i\,(t) = \textbf{output signal}$$

where $\alpha_i y_i(t) = f\,[\,\alpha_i x_i(t)\,]$

If the system not satisfies the superposition property, then the system is a nonlinear system .

Example : **Determine whether the following system is linear or nonlinear ?** $\mathbf{y(t) = |\,X(t)\,|}$

Solution: Input = $\alpha_1 x_1(t)$, produce output= $\alpha_1 y_1(t) = \alpha_1\,|\,X_1(t)\,|$

Input= $\alpha_2 x_2(t)$, produce output= $\alpha_2 y_2(t) = \alpha_2\,|\,X_2(t)\,|$.

Input= $\alpha_1 X_1(t) + \alpha_2 X_2(t)$, produce output = $|\,\alpha_1 X_1(t) + \alpha_2 X_2(t)\,|$

$\neq \alpha_1\,|\,X_1(t)\,| + \alpha_2\,|\,X_2(t)\,|$

$\neq \boldsymbol{\alpha_1 y_1(t) + \alpha_2 y_2(t)}$

∴ Superposition property not satisfies

∴ System is nonlinear.

Example: **Determine whether the following system is linear or nonlinear ?** $\mathbf{y(t) = x(t)\,Cos\,wt}$

Solution:

Input = $\alpha_1 x_1(t)$, produce output= $\alpha_1 y_1(t) = \alpha_1\ x_1 Cos\ wt$

Input= $\alpha_2 x_2(t)$, produce output= $\alpha_2 y_2(t) = \alpha_2\ x_2 Cos\ wt$.

Input= $\alpha_1 X_1(t) + \alpha_2 X_2(t)$, produce output = $(\alpha_1 X_1 + \alpha_2 X_2)\ cos\ wt$

$= \alpha_1\ X_1\ cos\ wt + \alpha_2\ X_2\ coswt$

$= \boldsymbol{\alpha_1 y_1(t) + \alpha_2 y_2(t)}$

Superposition property satisfies

∴ System is linear.

Example: **Determine whether the following system is linear or nonlinear? y(n)= 2 x(n)+3 where $x_1(n)=2$, $x_2(n)=3$**

at $x_1(n)$, $y_1(n) = 2\ x_1(n) + 3 = 2 \times 2 + 3 = 7$

at $x_2(n)$, $y_2(n) = 2\ x_2(n) + 3 = 2 \times 3 + 3 = 9$

at $x_3(n) = x_1(n) + x_2(n)$

$y_3(n) = 2\,[x_l(n) + x_2(n)] + 3 = 2\,[2 + 3\,] + 3 = 13 \neq yI\{n) + y2(n)$ ∴ system is nonlinear.

2.6.3. Static and Dynamic systems

Static system means instantaneous system; i.e. the response (output) at time ʎ depends only upon excitation (input) at lime ʎ, and independent on the future (t > ʎ)or past (t< ʎ) values of the excitation i.e. independent of the input applied before ʎ or after ʎ. In this case the system is called a zero memory or memoryless system. The static system contains no energy storage elements; i.e. the equation relating the output signal with respect to the input signal not contains derivative, integrals or time delays. If the previous characteristic not satisfies then the system is dynamic. This means that the dynamic system have a memory and its output depends on the past and present of the input signal in addition to the present input and contains derivatives, integrals and time delays. For convenience, every system will be assumed to have memory unless it is explicit stated as a memoryless system.

2.6.4. Time invariant system (fixed systems)

If the characteristics or properties of a system do not change with time, then the system is called a time invariant system. Otherwise the system is time varying. For time invariant system consider y(t) = sin u(t) , if u (0) = 1 ,then y(0) = sin 1.
System is time invariant because the output is always sin1 and no matter what t is. If a system is given by y(t) = (Sin t) u(t) , then when u(t) = 1 at t = 0, the output is y(0) = Sin (0) x l = 0 . If the same input is applied at t = 1 , then y(1) = (Sin 1) x 1 = 0.84 x 1 = 0.84 . We see that the output is different if the same input is applied at a different time. Thus the system described by
y(t) = (Sin t) u(t) is time varying system. Thus the time invariant system is given by y(t) = a u(t), where a = constant and the time varying system is given by yet) = a (t) u(t), where a(t) = time varying.

<u>Time Response of linear time invariant and continuous system</u>

We will develop a mathematical description for continuous time system. The description will be obtained from a special input - output pair of the system i.e. unit impulse input, unit step function or unit ramp function. The unit impulse cannot be generated in practice. Then we will calculate the impulse response without generating.

<u>The convolution integrals</u>

One of the most problems in system analysis is determining the output or response of a system to some specified input. The methods widely used are:-

1 - Solution of the system differential equations by classical methods.
2 - By using elementary signals and the principle of superposition property.
3 - By using transform methods. _

The superposition property is considered by the use of a technique known as convolution integrals as follows:-
Then the output of the system is h(t) or h (t-ʎ) , this output is known as impulse response of the system.
The impulse response h(t) or h (t-ʎ) can be used with a special input signal x(t) to calculate the output response y(t) for L.T.L system by using the convolution integral as follows:

$$y(t) = x(t)*h(t) = \int_{-\infty}^{\infty} x(ʎ)h(t-ʎ)dʎ = \int_{-\infty}^{\infty} h(ʎ)x(t-ʎ)dʎ = convolution\ integertal$$

This equation relates the input x(t) and output y(t) of a L.T.L system by means of the system impulse response h(t). .

Now we will study the different cases for the convolution integral:

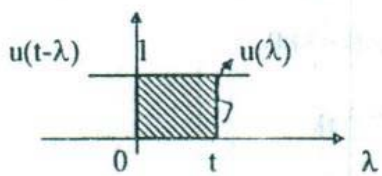
a) If x(t) & h(t) are causal

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\lambda)u(\lambda)h(t-\lambda)u(t-\lambda)d\lambda$$

OR

$$y(t) = \int_{-\infty}^{\infty} h(\lambda)u(\lambda)x(t-\lambda)U(t-\lambda)d\lambda$$

$$u(\lambda)u(t-\lambda) = \begin{cases} 1 & 0 \le \lambda < t \\ 0 & \text{otherwise} \end{cases}$$

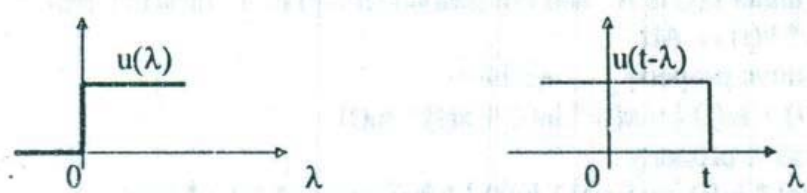


$$\therefore\ y(t) = \int_{0}^{t} x(\lambda)h(t-\lambda)d\lambda = \int_{0}^{t} h(\lambda)x(t-\lambda)d\lambda$$

b) If x(t) is causal and h(t) noncausal

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\lambda)u(\lambda)h(t-\lambda)d\lambda$$

$$= \int_{-\infty}^{\infty} h(\lambda)x(t-\lambda)u(t-\lambda)d\lambda$$



$$\therefore\ y(t) = \int_{0}^{t} x(\lambda)h(t-\lambda)d\lambda = \int_{-\infty}^{t} h(\lambda)x(t-\lambda)d\lambda$$

c) If x(t) is noncausal and h(t) causal

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\lambda)h(t-\lambda)u(t-\lambda)d\lambda = \int_{-\infty}^{\infty} h(\lambda)u(\lambda)x(t-\lambda)d\lambda$$

$$= \int_{-\infty}^{t} x(\lambda)h(t-\lambda)d\lambda = \int_{0}^{\infty} h(\lambda)x(t-\lambda)d\lambda$$

d) Integrate the multiplication to obtain the convolution integral. The following example illustrates the graphical solution.

consider $x3(t) = \int_{-\infty}^{\infty} X_1(\lambda) X_2(t-\lambda) d\lambda$

The area under X3(t) is iven by

$$\int_{-\infty}^{\infty} X_3(t)dt = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} X_1(\lambda) X_2(t-\lambda) d\lambda dt$$

$$= \int_{-\infty}^{\infty} X_1(\lambda)\left[\int_{-\infty}^{\infty} X_2(t-\lambda)dt\right]d\lambda$$

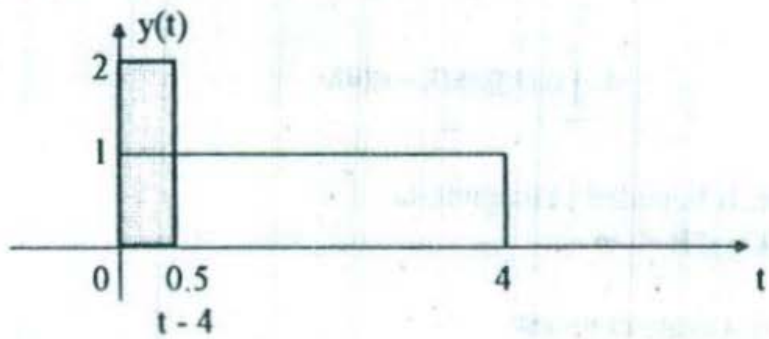
$$= \int_{-\infty}^{\infty} X_1(\lambda)\left[\text{area} - \text{under} - X_2(t)\right]d\lambda$$

$\therefore$ area under X3(t) = (area under X1(t)) × (area under X2(t))

BUT

$$y(t) = \int_{-\infty}^{\infty} h(\lambda) X(t-\lambda) d\lambda$$

$= h(\lambda)\ X(t\text{-}\lambda)$ - [width]



$$y(t) = \int_{0}^{t-4} 2.1\, d\lambda = 2\,\lambda\,|_{\lambda=0}^{\lambda=t-4} = 2\,(t-4)$$

at t = 4.5 $\therefore$ y(t) = 2 (4.5 - 4) = 2 × 0.5 = 1

at t = 5 $\therefore$ y(t) = 2 (5 - 4) = 2

or

<u>at t = 4.5</u>

$y(t) = h(\lambda)\ X(t\text{-}\lambda)$ - width = 1 × 2 × 0.5 = 1

<u>at t = 6.5</u>

$$y(t) = \int_{t-6}^{t-4} 2 \times 1 d\lambda = 2\lambda| = 4$$

$y(t) = h(\lambda)\ X(t\text{-}\lambda)$. width

= 1 × 2 . 2 = 4

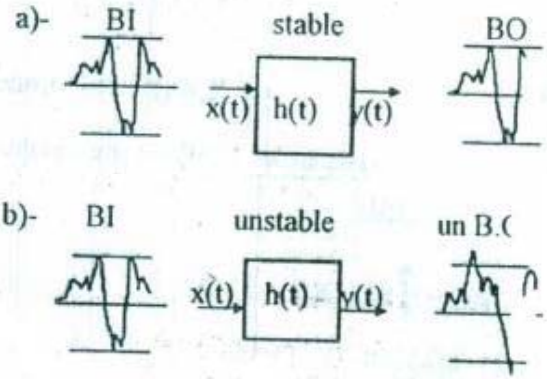
area = height of $h(\lambda)$ × height of $X(t\text{-}\lambda)$ × width

Stability of linear systems

A system is bounded input - bounded output (BIBO) stable if and only if every bounded input produces a bounded output as shown in Fig. (a). To derive this condition let us consider a fixed , linear system described by :

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)h(\lambda - t)d\lambda$$

$$\therefore |y(t)| = \left| \int_{-\infty}^{\infty} x(\lambda)h(\lambda - t)d\lambda \right|$$



The magnitude of an integral can never be greater than the integral of the magnitude ; i.e.

$$\left| \int_{-\infty}^{\infty} x(\lambda)h(\lambda - t)d\lambda \right| \le \int_{-\infty}^{\infty} |x(\lambda)h(\lambda - t)|d\lambda$$

$$\le \int_{-\infty}^{\infty} |x(\lambda)|.|h(\lambda - t)|d\lambda$$

If the input is bounded in magnitude ; i.e.

$$|x(\lambda)| \le R < \infty$$

where R is a finite constant

$$\therefore |y(t)| \le R \int_{-\infty}^{\infty} |h(t - \lambda)|d\lambda$$

let $\eta = t - \lambda$

$$|y(t)| \le R \int_{-\infty}^{\infty} |h(\eta)|d\eta$$

Thus the output y(t) is bounded if the following condition is satisfies ; i.e.

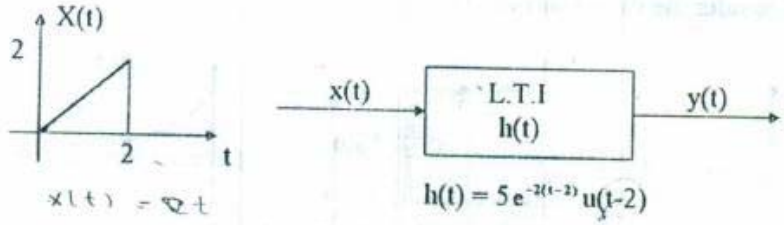
$$\int_{-\infty}^{\infty} |h(t)|dt < \infty \text{ ; i.e } \int_{-\infty}^{\infty} |h(t)|dt = \text{nonnegative and} < \infty$$

That is the area under the magnitude of the system impulse response be finite and nonnegative .

If h(t) starts from zero ; i.e. $t \ge 0$, then

$$\int_{0}^{\infty} |h(t)|\,dt = \text{nonnegative finite value} < \infty$$

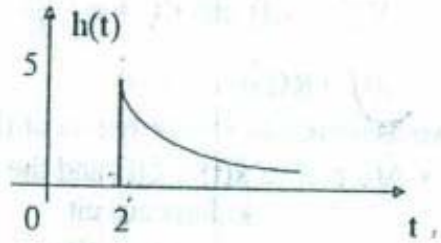
Example



Is the system BIBO. Stable ?

Solution

h(t) is shown in fig.( )



$$\int_{2}^{\infty} |h(t)|dt = \int_{2}^{\infty} 5e^{-2t}dt$$

$$= 5\left.\frac{e^{-2t}}{-2}\right|_{2}^{\infty}$$

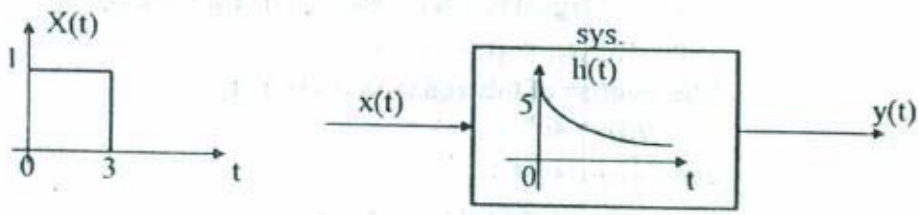
$$= 0 + \frac{5}{2}e^{-4}$$

$$= \frac{5}{2}e^{-4} < \infty$$

So the system is BIBO stable .

Example

$h(t) = 5\ e^{-2t}u(t)$ x(t) is shown in fig.



Is the system BIBO. Stable ?

SOL :

h(t) start at zero .

$$\therefore \int_{-\infty}^{\infty} |h(t)|dt = \int_{0}^{\infty} 5e^{-2t}dt$$

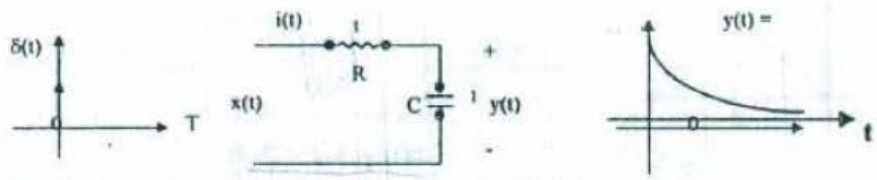
$$= -\left.\frac{5}{2}e^{-2t}\right|_{0}^{\infty}$$

$$= \frac{5}{2} < \infty$$

$\therefore$ system BIBO . stable .

1- Impulse Response of time invariant systems :

Consider the circuit shown in fig. ( )



The input voltage $x(t) = \delta(t)$, and the output is $y(t)$.
The circuit mathematical model is given by :

$$x(t) = I(t) \cdot R + V(t)$$

$$V(t) = y(t) \text{ and } i(t) = c\,\frac{dy(t)}{dt} = c\,\dot{y}(t)$$

$$\therefore x(t) = RC\,\dot{y}(t) + y(t)$$

we assume that the system is initially relaxed before application of the impulse .

- At $t \le 0^-$: $x(t) = \delta(t)$ and the capacitor is unchanged ; i.e. it represents a short circuit

$$\therefore \tau(0) = \frac{\delta(t)}{R} = \text{initial current}$$

- At $0^- < t \le 0^+$: the capacitor begin to carge by the initial current flow in C.

$$q(0) = \int_{0^-}^{0^+} i(o)dt = h\int_{0^-}^{0^+} \frac{s(t)}{R}dt = \frac{1}{R}\int_{0^-}^{0^+} S(t)dt = \frac{1}{R}$$

then the initial voltage $Vc(o) = y(0) = \frac{q(0^+)}{c} = \frac{1}{RC}$

- At $t > 0^+$: input signal $\delta(t) = 0$; i.e. input is short circuit , and the capacitc is discharge through R. The circuit equation will be

$$0 = RC\,\dot{y}(t) + y(t)$$

The solution of this equation is given by :

$$y(t) = Ae^{pt}$$

and $\dot{y}(t) = APe^{pt}$

$$\therefore 0 = RC(APe^{pt}) + Ae^{pt}$$

$$= A(RCP + 1)e^{pt}$$

$$\therefore RCP + 1 = 0$$

$$\therefore p = -\frac{1}{RC}$$

$$\therefore y(t) = Ae^{pt}$$

At $t = 0$ $\quad \therefore y(0) = A = \frac{1}{RC}$

$$\therefore y(t) = \frac{1}{RC}e^{-t/RC} = h(t) \quad \ldots\ldots \quad t \ge 0$$

$h(t)$ = impulse response of the system ; i.e. it is the response of the system to a unit impulse input applied at $t = 0$, with all initial conditions of the system equal zero .

$$\therefore h(t) = \frac{1}{RC}e^{-t/RC} \quad \ldots\ldots \quad t \ge 0$$

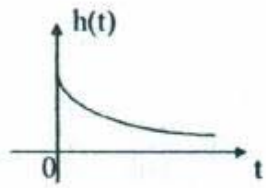
This means that when the system input is unity impulse δ(t) , then the impulse response of the system h(t) is equal to the system output y(t) .

$$h(t) = L^{-1}[H(s)] = F^{-1}[H(w)]$$

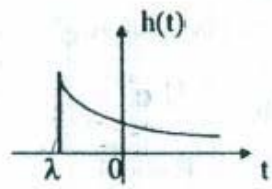
This means that h(t) is related the system input and output ; i.e. we apply the input δ(t) and measure the output y(t) . The output will be denoted by h(t) and will be called the contiuous time impulse response of the system [h(t)] .

If the system is causal , this means no output will apear before the application of the impulse δ(t)

$$\therefore h(t) = 0 \quad , \quad \text{for} \quad t < 0$$



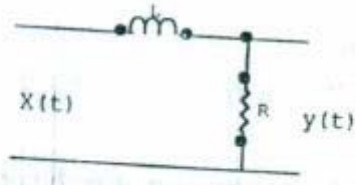
$h(t) = 0 \; . \; t < 0$
Causal system



$h(t) \neq 0 \; , \; t < 0$
noncausal system

Example

Ex : Find the impulse response of the system . shown in fig. ( )



Sol :

$$x(t) = \delta(t) \quad , \quad y(t) = h(t)$$

$$\delta(t) = V_L + V_R$$

$$= L\frac{di}{dt} + y(t)$$

$$y(t) = i(t)\,R \quad , \quad \frac{dy}{dt} = \frac{di}{dt}R$$

$$\therefore \frac{di}{dt} = \frac{1}{R}\frac{dy(t)}{dt}$$

$$\therefore \delta(t) = \frac{L}{R}\frac{dh(t)}{dt} + h(t)$$

For $t < 0$, $\delta(t) = 0$

$$\therefore 0 = \frac{L}{R}\dot{h}(t) + h(t)$$

→ solution of diff. Eq. is

$$h(t) = Ae^{pt}$$

$$\dot{h}(t) = PAe^{pt}$$

$$\therefore 0 = (L/R)\,PA\,e^{pt} + A\,e^{pt}$$

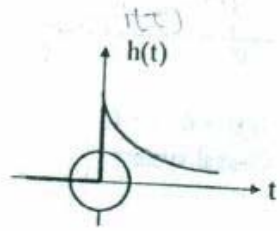
$$0 = A\left(\frac{L}{R}P + 1\right)e^{pt}$$

$$\therefore \frac{L}{R}P = -1 \quad \therefore P = -\frac{R}{L}$$

$$h(t) = A\,e^{-Rt/L}$$

$$\therefore \frac{L}{R}\dot{h} + h(t) = \delta(t)$$

$$\int_{0^-}^{0^+}\frac{L}{R}\dot{h} + \int_{0^-}^{0^+}h(t) = \int_{0^-}^{0^+}\delta(t) = 1$$



$\therefore$ h(t) is discontinous at $t = 0$

$$\therefore \int_{0^-}^{0^+}h(t) = 0 \quad , \quad h(0^-) = 0$$

$$\int_{0^-}^{0^+}\frac{L}{R}\dot{h} = 1$$

$$(L/R)\,[h(0^+) - \cancel{h(0^-)}] = 1 \quad \text{zero}$$

$$h(0^+) = R/L = A \qquad\qquad h(t) = \frac{R}{L}e^{-Rt/L}$$

Ex: Find the impulse response of a system modeled by the differetial equation $\tau_0 \dfrac{dy(t)}{dt} + y(t) = X(t) \qquad \infty < t < \infty$

where X(t) is the input and y(t) the output .

Sol :

The impulse response h(t) of a fixed , linear system assumed initially unexcited ; i.e. Zero initial condition , to be the response of the system to a unit impulse applied at t = 0 .

Setting $x(t) = \delta(t)$ results in the response $y(t) = h(t)$.

For $t > 0$, $\delta(t) = 0$, the differential equation for the impulse response is given by :

$$\tau_0 \frac{dh(t)}{dt} + h(t) = 0 \qquad t > 0$$

Assuming a solution of the form

$$h(t) = A.e^{Pt}$$

$$\& \quad h(t) = PAe^{Pt}$$

$$\tau_0 PAe^{Pt} + Ae^{Pt} = 0$$

$$\therefore (\tau_0 P + 1) Ae^{Pt} = 0$$

$$\tau_0 P + 1 = 0$$

$$P = -1 / \tau_0$$

$$\therefore h(t) = Ae^{-t/\tau_0} \qquad t > 0$$

To find A , we require an initial condition for h(t) . The system is unexcited for $t < 0$

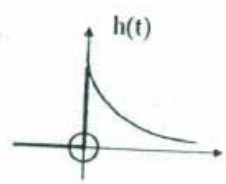
$$\therefore h(t) = 0 \qquad t < 0$$

For $t \geq 0$

$$\tau_0 \frac{dh(t)}{dt} + h(t) = \delta(t) \qquad t \geq 0$$

The integral of this equation from $t = 0^-$ to $t = 0^+$ gives

$$\int_{0^-}^{0^+} \tau_0 \frac{dh(t)}{dt} + \int_{0^-}^{0^+} h(t)dt = \int_{0^-}^{0^+} \delta(t)dt = 1$$



h (t) is discontinuous at t = 0

$$\therefore \int_{0^-}^{0^+} h(t)dt = 0$$

$$\int_{0^-}^{0^+} \tau_0 \frac{dh(t)}{dt} = \tau_0 [\, h(0^+) - h(0^-) \,] + 0 = 1$$

since $h(0^-) = 0$

$$\therefore h(0^+) = 1 / \tau_0$$

$$\therefore h(0^+) = A = 1 / \tau_0$$

$$\therefore h(t) = \begin{cases} 0 & t < 0 \\ \dfrac{1}{\tau_0} e^{-t/\tau_0} & t \geq 0 \end{cases}$$

Where h(t) is the impulse response of the system .